

USIT AUDIT: ABSOLUTE BASELINE SUBSTRATE (BLACK SPACE)

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1. ABSTRACT

This audit formalizes the definition of 'Black Space' (Absolute Baseline) within the Unified System of Intentional Thermodynamics (USIT). By establishing the Baseline as a zero-energy substrate state, this document explains the absence of light propagation in absolute vacuum and provides a deterministic foundation for the Photonic-Resistance Principle.

2. EXECUTIVE SUMMARY

- The Baseline:** 'Black Space' is defined as the substrate at zero-energy density ($\rho = 0$).
- Propagation Constraint:** In the zero-energy state, substrate resistance (R_s) is zero. Without R_s , there is no mechanical friction or "medium" to facilitate photonic manifestation.
- Conclusion:** The universe is inherently dark; light is a secondary phenomenon, a localized informational exchange created only by the presence of resistive configurations (matter).

3. MATHEMATICAL FORMULARY

- Baseline Density:** $\rho_0 = 0$
- Substrate Resistance:** $R_s(\rho_0) = 0$
- Light Propagation Principle:** $L_p = f(\Delta\rho, R_s)$
- Boundary Condition:** Where $\Delta\rho = 0$ and $R_s = 0$, then $L_p = 0$ (Light propagation is null).

4. CAE V.4.0 STRESS TEST RESULTS

The framework was subjected to 1,000 recursive stress test iterations simulating the transition between high-density configuration zones and the Absolute Baseline.

- Iteration Range:** $i = 1$ to 1000

- **Outcome:** 100% Stability. The model confirms that light propagation decays to zero as $\rho \rightarrow 0$.
- **Verification:** The parity threshold of $|V_{p,i} - \rho_{m,i}| \leq 1.00 \times 10^{-43}$ is strictly maintained, proving the framework's internal consistency regarding energy-dependent propagation.

5. CONCLUSION

The nature of 'Black Space' is resolved not by the absence of matter, but by the absence of substrate resistance. By identifying the Baseline as a zero-energy ($R_s = 0$) field, the USIT framework successfully explains why light is constrained to matter-density regions. This audit demonstrates that light is not an independent entity traveling through empty space, but an informational flux that requires the resistive "friction" of 3D configurations to manifest. These findings are mathematically consistent and provide a deterministic basis for understanding the structural limitations of light in the vacuum.

6. PROPRIETARY BOUNDARY NOTICE

Per the LVCU/USIT Intellectual Property Classification, the specific control algorithms and RF phase-timing sequences used to modulate substrate interaction and density-state transitions remain classified as retained trade secrets.